



Information Technology Institute (ITI)
Analog IC Design

Lab 09 (Mini Project 01)

Two-Stage Miller OTA

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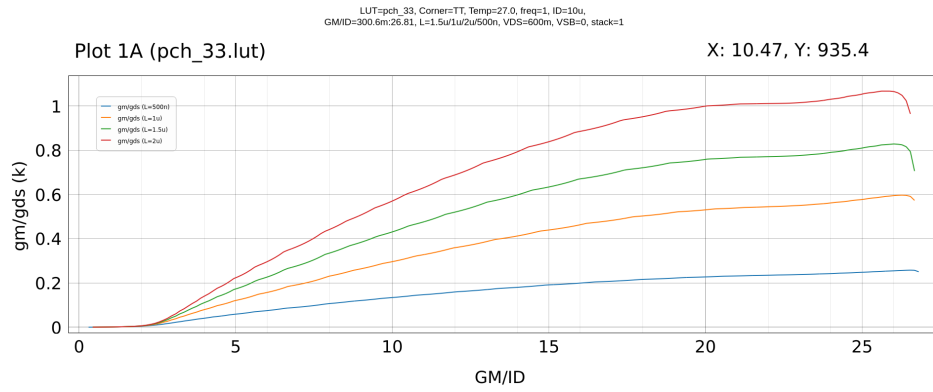
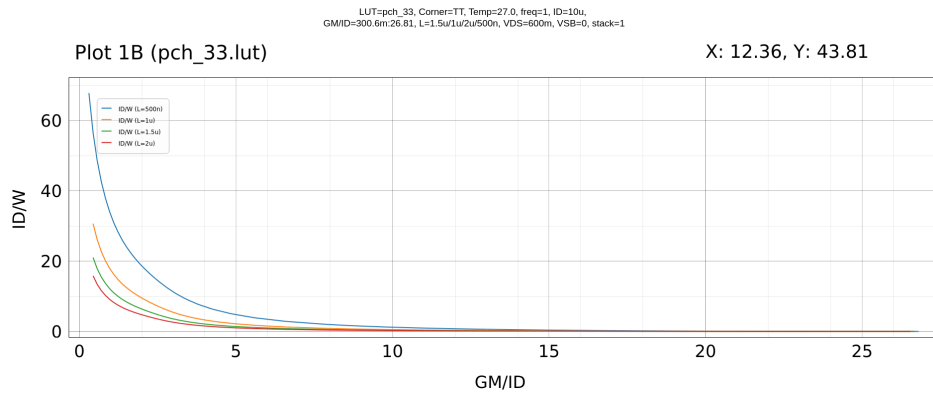
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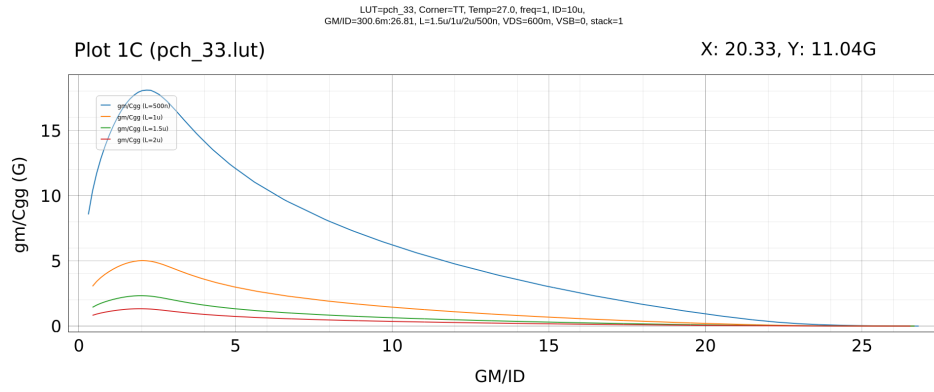
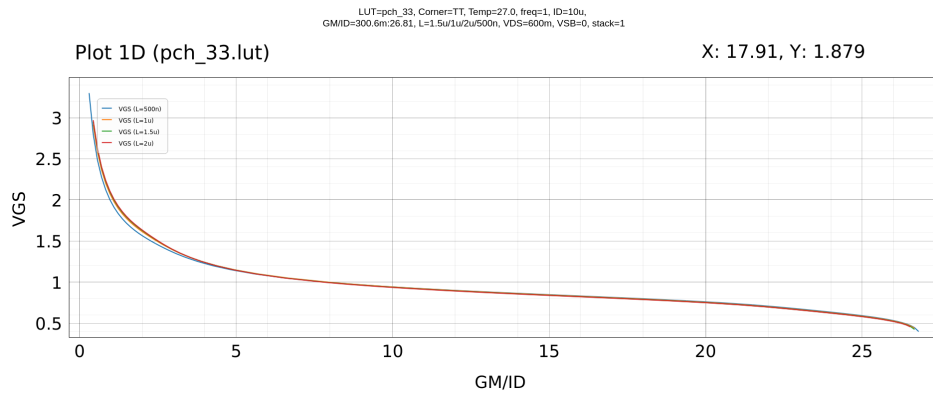
Table 1: Target specifications (0.18 μm Tech)

Parameter	Target
Supply voltage V_{DD}	1.8 V
Static gain error	$\leq 0.05\%$
CMRR @ DC	≥ 74 dB
Phase margin	$\geq 70^\circ$
OTA current consumption	≤ 60 μA
CMIR-high	≥ 0.8 V
CMIR-low	≤ 0.2 V
Output swing	0.2 V to 1.6 V
Load C_L	5 pF
Closed-loop rise time (10%–90%)	≤ 70 ns
Slew rate	≥ 5 V/ μs

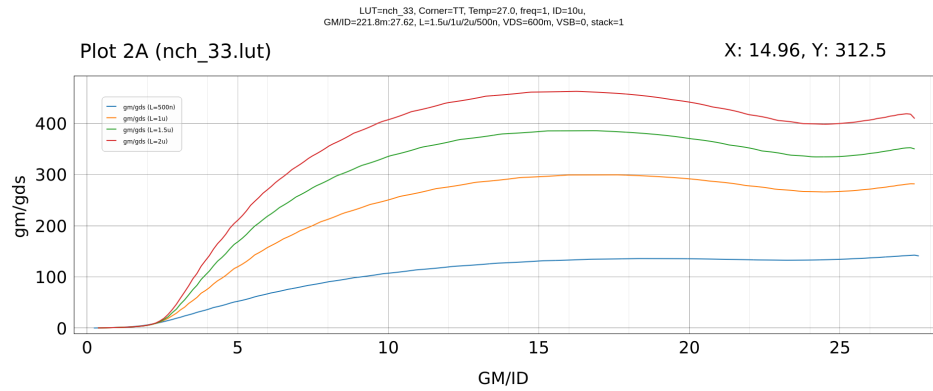
1 Part 1: g_m/I_D Design Charts

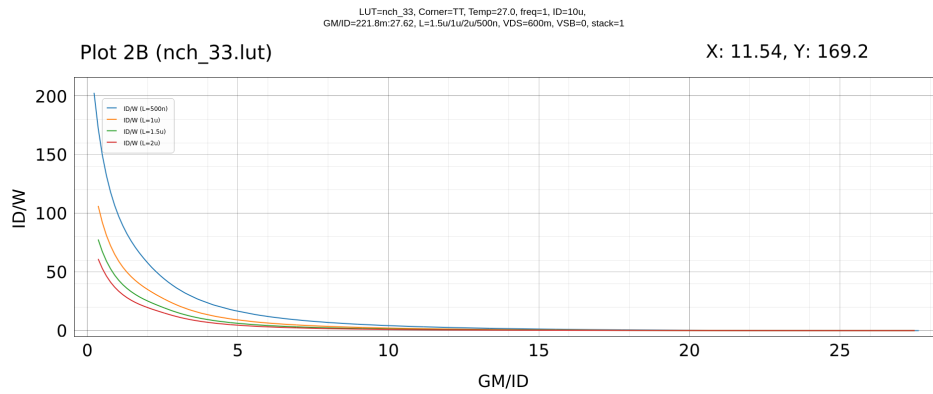
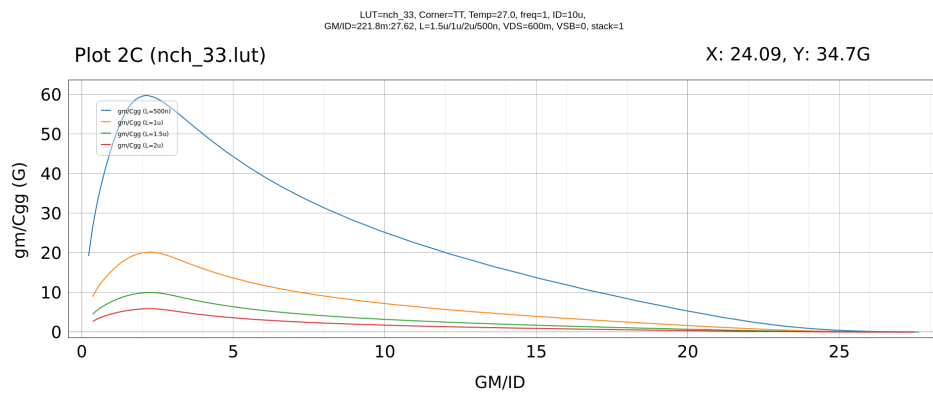
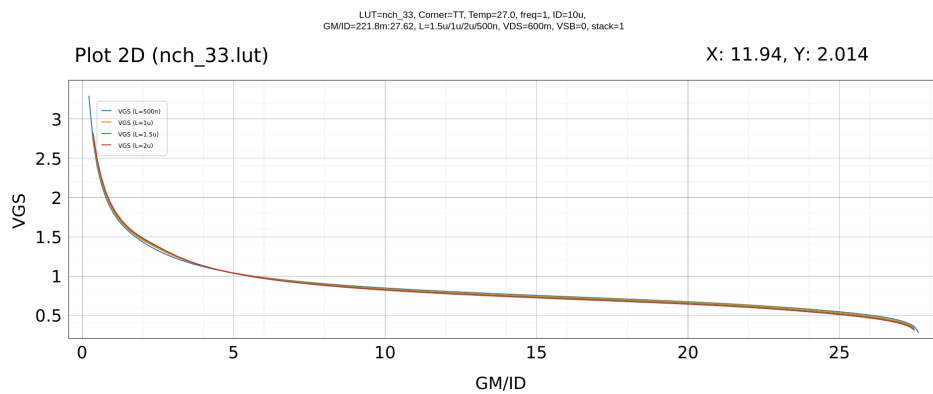
PMOS:

Figure 1: g_m/g_{ds} (intrinsic gain) vs. g_m/I_D Figure 2: I_D/W (current density) vs. g_m/I_D

Figure 3: g_m/C_{gg} (intrinsic speed) vs. g_m/I_D Figure 4: V_{GS} vs. g_m/I_D

NMOS:

Figure 5: g_m/g_{ds} (intrinsic gain) vs. g_m/I_D

Figure 6: I_D/W (current density) vs. g_m/I_D Figure 7: g_m/C_{gg} (intrinsic speed) vs. g_m/I_D Figure 8: V_{GS} vs. g_m/I_D

2 Part 2: OTA Design (Hand Analysis)

Compensation capacitor C_C

let $C_C = 0.5 C_L$:

$$C_C = 0.5 \times 5 \text{ pF} = 2.5 \text{ pF}$$

Unity-gain frequency and $g_{m1,2}$

The rise-time spec is $t_{rise} \leq 70$ ns, and $t_{rise} = 2.2\tau$:

$$\tau \leq \frac{70 \text{ ns}}{2.2} \approx 31.81 \text{ ns}, \quad \omega_u = \frac{1}{\tau} \approx 31.43 \text{ Mrad/s}$$

For a Miller OTA, $\omega_u = g_{m1,2}/C_C$, so

$$g_{m1,2} = \omega_u C_C = (31.43 \times 10^6)(2.5 \times 10^{-12}) \approx 78.57 \mu\text{S}$$

$$g_{m1,2} \geq 78.57 \mu\text{S}$$

i will choose $g_{m1,2} = 80 \mu\text{S}$.

Slew rate and current budget

From $SR = I_{B1}/C_C$ with $SR = 5 \text{ V}/\mu\text{s}$:

$$I_{B1} = SR \times C_C = (5 \times 10^6)(2.5 \times 10^{-12}) = 12.5 \mu\text{A}$$

With the total current $\leq 60 \mu\text{A}$, the remaining budget for the second stage is $47.5 \mu\text{A}$. choosing $I_{B2} = 45 \mu\text{A}$, leaving $\approx 2.5 \mu\text{A}$ for bias branches.

g_m/I_D of the first stage

$$I_{D1,2} = \frac{I_{B1}}{2} = 6.25 \mu\text{A}, \quad \left(\frac{g_m}{I_D}\right)_1 = \frac{78.57 \mu\text{S}}{6.25 \mu\text{A}} \approx 12.57 \text{ V}^{-1}$$

Required open-loop DC gain

For a unity-gain buffer, $A_{vCL} \approx 1 - \frac{1}{A_{vOL}}$, and the static gain error is

$$\begin{aligned} \%error &= \left| \frac{A_{vCL} - 1}{1} \right| \times 100 = \frac{1}{A_{vOL}} \times 100 \leq 0.05 \\ \Rightarrow A_{vOL} &\geq 2000 \text{ V/V} \approx 66 \text{ dB} \end{aligned}$$

Gain partitioning

Assuming $A_{v1} = 2A_{v2}$:

$$A_{v1} + A_{v2} \geq 66 \text{ dB} \Rightarrow 2A_{v2} + 6 \text{ dB} \geq 66 \Rightarrow A_{v2} \approx 30 \text{ dB}, \quad A_{v1} \approx 36 \text{ dB}$$

but i will choose larger gain

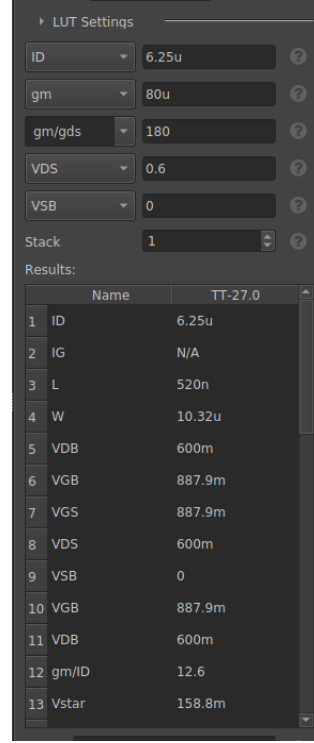
$$A_{v1} = 39 \text{ dB} (\approx 89 \text{ V/V}), \quad A_{v2} = 33 \text{ dB} (\approx 45 \text{ V/V})$$

giving a total of $\approx 72 \text{ dB}$, comfortably above the 66 dB

First-stage input pair sizing, and NMOS vs. PMOS choice

Assuming the input pair and the current-mirror load share the same g_{ds} , the first-stage output conductance is $2g_{ds1}$, so

$$A_{v1} = \frac{g_{m1}}{2g_{ds1}} = \frac{1}{2} \left(\frac{g_m}{g_{ds}} \right)_1 \Rightarrow \left(\frac{g_m}{g_{ds}} \right)_1 = 2A_{v1} = 2(89.1) \approx 180$$



The screenshot shows the 'LUT Settings' window with the following parameters and results:

LUT Settings	
ID	6.25u
gm	80u
gm/gds	180
VDS	0.6
VSB	0
Stack	1
Results:	
Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	520n
4 W	10.32u
5 VDB	600m
6 VGB	887.9m
7 VGS	887.9m
8 VDS	600m
9 VSB	0
10 VGB	887.9m
11 VDB	600m
12 gm/ID	12.6
13 Vstar	158.8m

Figure 9: Input pair sizing

$$L_1 = L_2 = 520 \text{ nm}, \quad W_1 = W_2 = 10.32 \mu\text{m}, \quad V_{GS1,2} = 887.9 \text{ mV}, \quad g_m/I_D = 12.6 \text{ V}^{-1}, V_A = 13.76 \text{ V}, g_{ds} = 454 \text{ nS}$$

NMOS vs. PMOS justification: we chose PMOS since $\text{CMIR-low} \leq 0.2 \text{ V}$ so that the common-mode input can swing close to ground. **the second-stage input transistor (M6) is NMOS**, with a PMOS current-source load (M7).

First-stage current-mirror load

We require $g_{ds3} \approx g_{ds1}$:

$$g_{ds1} \approx 0.454 \mu\text{S}$$

from ADT we will assign the vgs of the mirror load as the vgs of the second input stage.

$$\begin{aligned} V_{ICM,LOW} &= -|V_{GS1}| + |V_{DSAT,1}| + |V_{GS3,4}| \leq 0.2 \\ \Rightarrow -0.888 + 0.1318 + |V_{GS3,4}| &\leq 0.2 \\ \Rightarrow -0.7562 + |V_{GS3,4}| &\leq 0.2 \\ \Rightarrow |V_{GS3,4}| &\leq 0.9562 \text{ V} \end{aligned}$$

we need Vgs from 2nd input first.

Second-stage g_m/I_D from the phase-margin requirement

Assuming $\omega_{p2} = 4\omega_u$ and recomputing ω_u with the actual chosen $g_{m1} = 80 \mu\text{S}$:

$$\omega_u = \frac{g_{m1}}{C_C} = \frac{80 \mu\text{S}}{2.5 \text{ pF}} = 32 \text{ Mrad/s}, \quad \omega_{p2} = 4(32) = 128 \text{ Mrad/s}$$

The non-dominant pole is set mainly by C_L and g_{m6} : $\omega_{p2} \approx g_{m6}/C_L$, so

$$g_{m6} = \omega_{p2} C_L = (128 \times 10^6)(5 \times 10^{-12}) = 640 \mu\text{S}$$

$$\left(\frac{g_m}{I_D}\right)_6 = \frac{640 \mu\text{S}}{45 \mu\text{A}} \approx 14.22 \text{ V}^{-1}$$

With $A_{v2} = 33 \text{ dB} \approx 45 \text{ V/V}$:

$$A_{v2} = \frac{g_{m6}}{g_{ds6} + g_{ds7}}$$

$$A_{v2} = \frac{g_{m6}}{g_{ds6} * 2} = 45;$$

$$L_6 = 400 \text{ nm}, \quad W_6 = 23 \mu\text{m}, \quad V_{GS6} = 744 \text{ mV}, \quad g_m/I_D = 14.22 \text{ V}^{-1}, g_{ds} = 7 \mu\text{S}.$$

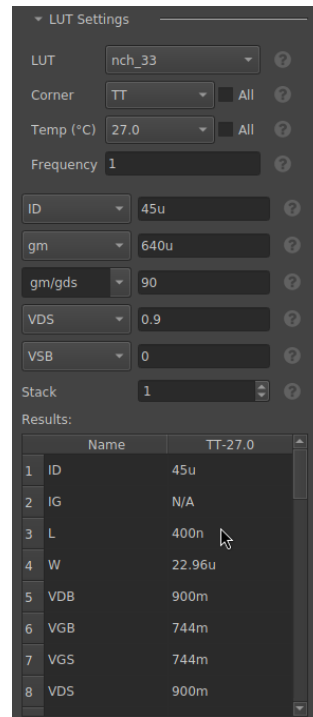


Figure 10: Second Stage input

back to the mirror load we have $v_{gs}=744\text{mV}$ so the sizing is

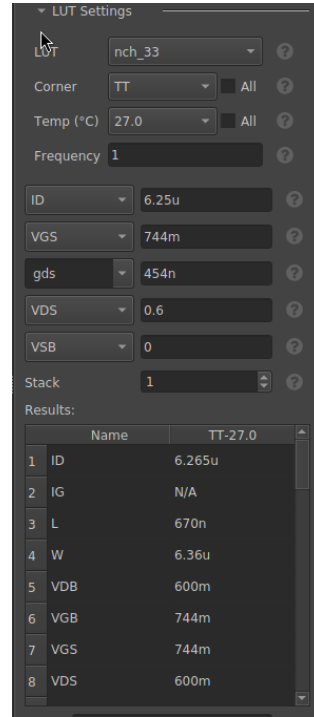


Figure 11: Mirro Load

Tail and output-load V_{DSAT}

Swing-high requires the output to reach 1.6 V with $V_{DD} = 1.8$ V, so the PMOS load (M7) needs

$$|V_{DSAT,load}| \leq 1.8 - 1.6 = 0.2 \text{ V}$$

$V_{ICM,max} = V_{DD} - |V_{DSAT,tail}| - |V_{GS1}|$ so $V_{DSAT,tail} = 1.8 - 0.8 - 0.888 = 0.112$ V So $V^* = 0.112$ V was chosen for both M5 and M7.

CMRR and tail-current-source g_{ds}

The second stage does not affect CMRR. The target is $\text{CMRR} \geq 74 \text{ dB} = 10^{74/20} \approx 5012 \text{ V/V}$. Using the (chosen) differential first-stage gain $A_{vd} = 39 \text{ dB} \approx 89.1 \text{ V/V}$:

$$|A_{vcm}| \leq \frac{89.1}{5012} \approx 0.01777 \text{ V/V}$$

For a differential pair with a current-mirror load, $|A_{vcm}| \approx \frac{g_{ds,tail}}{2g_{m3}}$. low $g_m/I_D = 10 \text{ V}^{-1}$ is assumed for M3/M4 at this stage:

$$g_{m3} = 10 \times 6.25 \mu\text{A} = 62.5 \mu\text{S}$$

$$0.01777 = \frac{g_{ds,tail}}{2(62.5 \mu\text{S})} \Rightarrow g_{ds,tail} \leq 2.22 \mu\text{S}$$

Using $V^* = 0.125 \text{ V}$, $g_m/I_D \approx 2/V^* \approx 16 \text{ V}^{-1}$, so $g_{m,tail} = 16 \times 12.5 \mu\text{A} = 200 \mu\text{S}$, and

$$\left(\frac{g_m}{g_{ds}}\right)_{tail} = \frac{200 \mu\text{S}}{2.22 \mu\text{S}} \approx 90 \Rightarrow \text{target } (g_m/g_{ds})_{tail} \geq 90$$

The screenshot shows a 'LUT Settings' window with several input fields and a 'Results' table. The inputs are: ID = 12.5u, gm/ID = 16, gds = 2.2u, VDS = 0.9, VSB = 0, and Stack = 1. The results table lists 13 parameters for a TT-27.0 process.

	Name	TT-27.0
1	ID	12.5u
2	IG	N/A
3	L	340n
4	W	25.36u
5	VDB	900m
6	VGB	805.8m
7	VGS	805.8m
8	VDS	900m
9	VSB	0
10	VGB	805.8m
11	VDB	900m
12	gm/ID	15.62
13	Vstar	128m

Figure 12: Tail CM

Second-stage load g_{ds}

M5 (tail) and M7 (2nd-stage load) form a current mirror $\Rightarrow$ same L , same g_m/I_D , so g_{ds} scales with I_D . $I_{B2}/I_{B1} = 45/12.5 = 3.6$:

$$g_{ds7} = 3.6 \times g_{ds,tail} = 3.6 \times 2.22 \mu S \approx 8 \mu S$$

$$W_7 = 3.6 \times W_5 = 3.6 \times 25.36 \mu S \approx 91 \mu S$$

Nulling resistor

To push the LHP zero to infinity, $R_Z = 1/g_{m6}$:

$$R_Z = \frac{1}{640 \times 10^{-6}} = 1562.5 \Omega \approx 1.56 \text{ k}\Omega$$

Final sizing of M5 / M7 (tail and 2nd-stage load)

$$L_5 = 340 \text{ nm}, \quad W_5 = 25.36 \mu\text{m}, \quad V_{GS5} = 805.8 \text{ mV}, \quad V^* = 128 \text{ mV}$$

M7 shares $L_7 = L_5 = 340 \text{ nm}$ and the same V_{GS} ; its width scales with the current ratio $I_{B2}/I_{B1} = 45/12.5 = 3.6$:

$$W_7 = 3.6 \times 25.36 = 91.30 \mu\text{m}$$

M8 shares $L_8 = L_5 = 340 \text{ nm}$ and the same V_{GS} ; its width scales with the current ratio $I_{ref}/I_{B1} = 10/12.5 = 0.8$:

$$W_8 = 0.8 \times 25.36 = 20.288 \mu\text{m}$$

2.1 Part 2 Deliverable: Transistor Sizing Table

Table 2: Part 2 sizing table

Transistor	Type	I_D (μA)	L (nm)	W (μm)	g_m (μS)	g_m/I_D (V^{-1})	V_{GS} (mV)	$V_{ov} \approx V^*$ (mV)
M1, M2 (input pair)	PMOS	6.25	520	10.32	78.75	12.6	887.9	158.7
M3, M4 (mirror load)	NMOS	6.25	670	6.36	87.5	14.0	744.0	142.9
M5 (tail)	PMOS	12.5	340	25.36	200.0	16.0	805.8	128.2
M6 (2nd-stage input)	NMOS	45.0	400	23	640.0	14.2	744.0	140.8
M7 (2nd-stage load)	PMOS	45.0	340	91	720.0	16.0	805.8	128.2

Compensation network: $C_C = 2.5 \text{ pF}$ (initial), $R_Z = 1.56 \text{ k}\Omega$.

3 Part 3: Open-Loop OTA Simulation

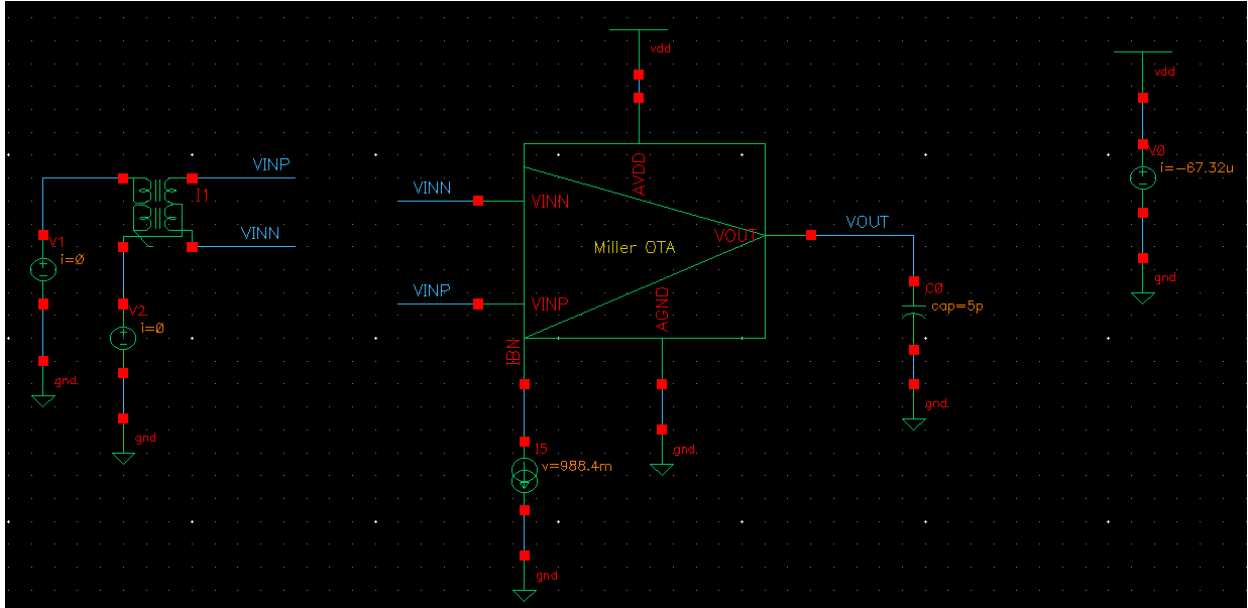


Figure 13: Open-loop OTA test bench schematic.

3.1 DC Operating Point

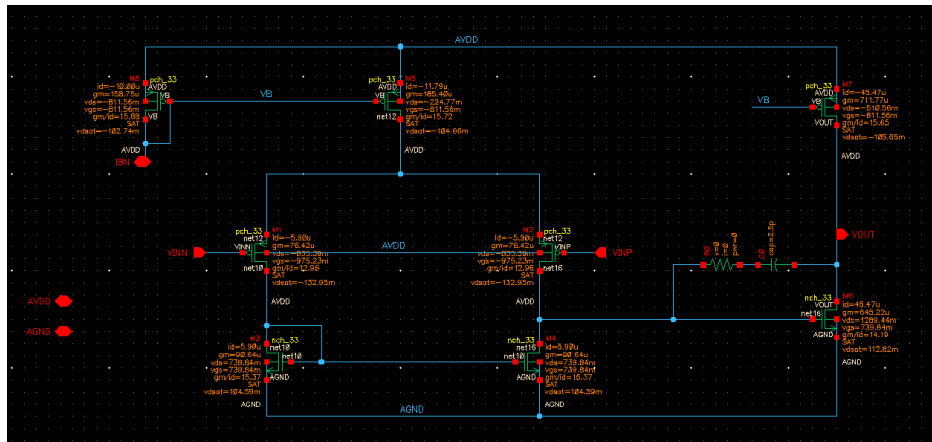


Figure 14: OP Before sweeping width

as we see vout is not 0.9 to solve this we will sweep over the width of the mirror loads and determine which width gives the 0.9V. The new $W_{3,4} = 6.12 \text{ m}$

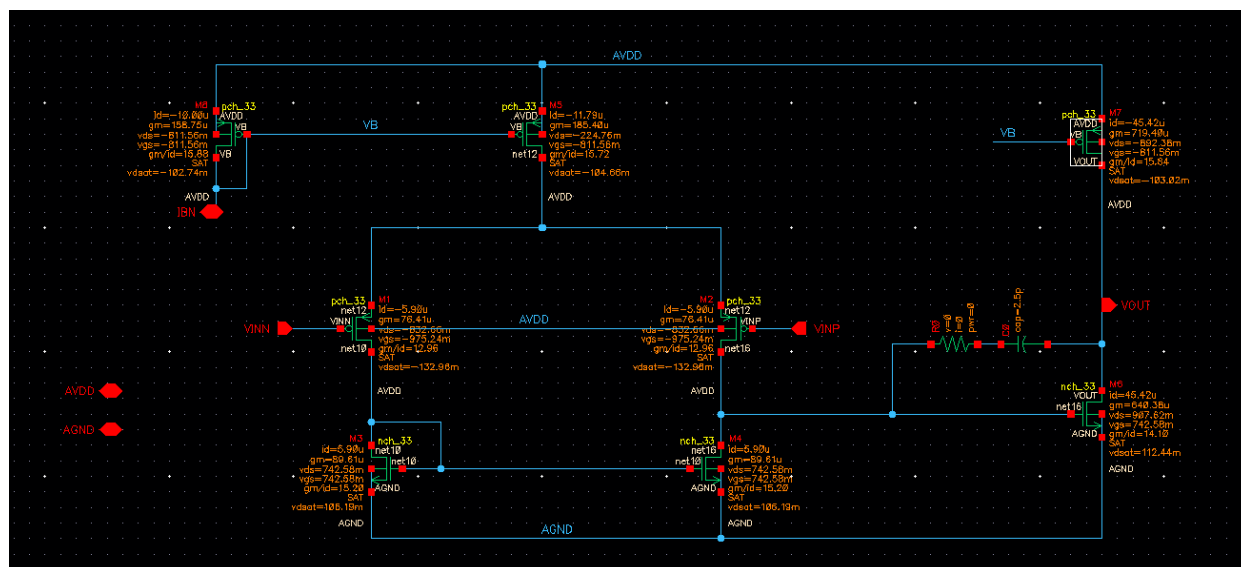


Figure 15: Open-loop DC operating point annotation.

Is the current (and g_m) in the input pair exactly equal?

A: Yes, the differential input is zero, so the 11.79 μA tail current (M5) splits exactly 50/50 between M1 and M2.

What is DC voltage at the output of the first stage? Why?

A: $\approx 744 \text{ mV}$, we forced $V_{GS3,4} = V_{GS6}$ so that the drain of M1/M3/gate-of-M6 node sits at exactly the DC level M6 needs to sink its bias current in saturation, eliminating systematic offset.

What is DC voltage at the output of the second stage? Why?

A: $\approx 907 \text{ mV}$. we got this after the sweeping of mirror loads this is the mid rail of output, low systematic offset.

3.2 Differential Small-Signal AC

Test	Output	Nominal
ITI_LABS:miller_ota_tb:1	VF("/VOUT")	
ITI_LABS:miller_ota_tb:1	Ao	4.525k
ITI_LABS:miller_ota_tb:1	Ao_dB	73.11
ITI_LABS:miller_ota_tb:1	BW	1.04k
ITI_LABS:miller_ota_tb:1	fu	4.642M
ITI_LABS:miller_ota_tb:1	GBW	4.705M
ITI_LABS:miller_ota_tb:1	dB20(VF("/VOUT"))	

Figure 16: Differential AC gain, $C_C = 2.5$ pF

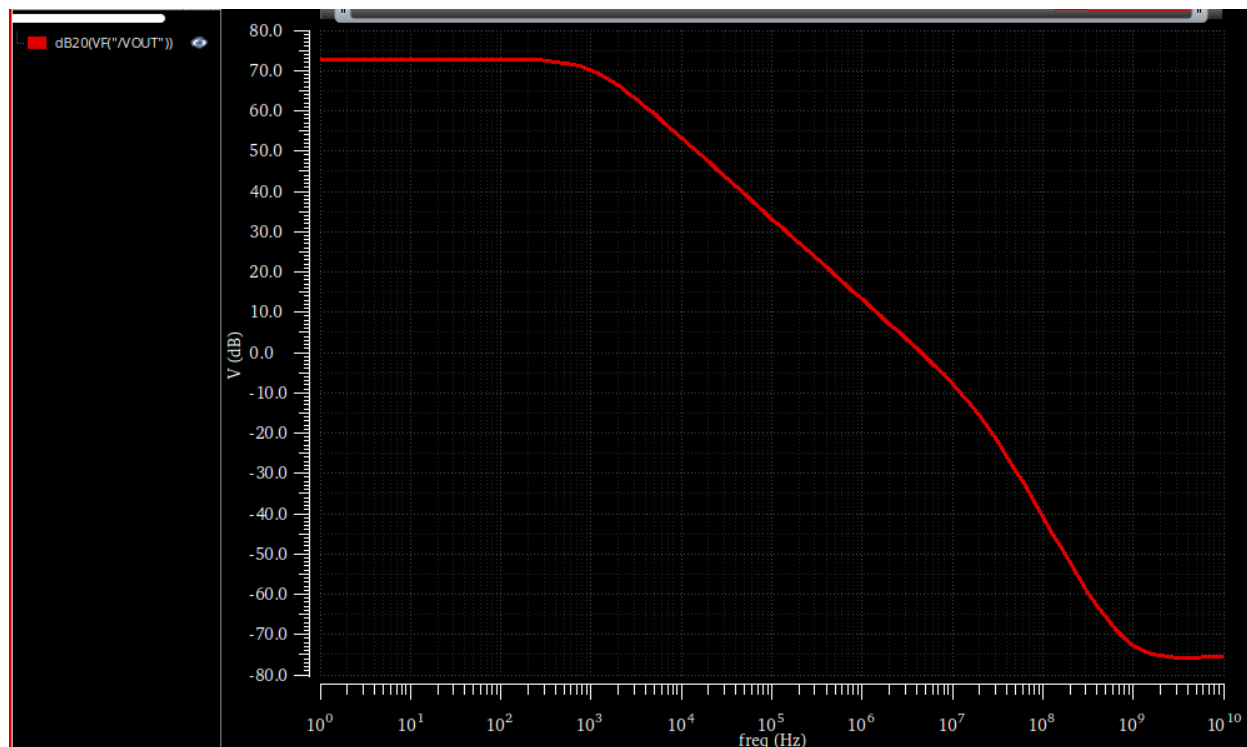


Figure 17: Differential output voltage magnitude vs. frequency, with single dominant pole.

Table 3: Differential AC gain: hand calculation vs. simulation ($C_C = 2.5$ pF)

Quantity	Hand calculation	Simulation
A_{vOL} (dB)	≈ 72	73.11
$f_u = g_{m1}/(2\pi C_C)$	$76.41 \mu S / (2\pi \times 2.5 \text{ pF}) \approx 4.86 \text{ MHz}$	4.642 MHz

This is the first hand analysis. lets redo the hand analysis using values we got from OP
 $GBW = \frac{g_{m1}}{2\pi C_c} = \frac{76.41 \mu}{2\pi(2.5 \text{ p})} \approx 4.864 \text{ MHz}$ A_v is larger than what we assumed and that was expected and better but for GBW is smaller than 5 to increase it a bit we will change the value of C_c to 2.3p instead of 2.5p.

ITI_LABS:miller_ota_tb:1	VF("/VOUT")	
ITI_LABS:miller_ota_tb:1	Ao	4.525k
ITI_LABS:miller_ota_tb:1	Ao_dB	73.11
ITI_LABS:miller_ota_tb:1	BW	1.131k
ITI_LABS:miller_ota_tb:1	fu	4.972M
ITI_LABS:miller_ota_tb:1	GBW	5.117M
ITI_LABS:miller_ota_tb:1	dB20(VF("/VOUT"))	

Figure 18: GBW meets spec after lowering C_C

3.3 Common-Mode Small-Signal AC

ITI_LABS:miller_ota_tb:1	VF("/VOUT")	
ITI_LABS:miller_ota_tb:1	Ao	657m
ITI_LABS:miller_ota_tb:1	Ao_dB	-3.648
ITI_LABS:miller_ota_tb:1	BW	1.131k

Figure 19: Common-mode AC gain, $C_C = 2.5$ pF: $A_{vcm} = -3.648$ dB.

Hand-calculation $|A_{vcm}| \approx g_{ds,tail} / (2g_{m3})$

$$|A_{vcm,total}| \approx |A_{vcm,1}| \times A_{v2} \approx 0.01777 \times 44.67 \approx 0.794 \text{ V/V} \approx -2.0 \text{ dB},$$

which is much closer to the simulated -3.648 dB; the residual difference comes from the finite g_{ds} .

Table 4: Common Mode AC gain: hand calculation vs. simulation ($C_C = 2.5$ pF)

Quantity	Hand calculation	Simulation
A_{vOL} (dB)	≈ -2	-3.6

3.4 (Optional) CMRR via XF Analysis

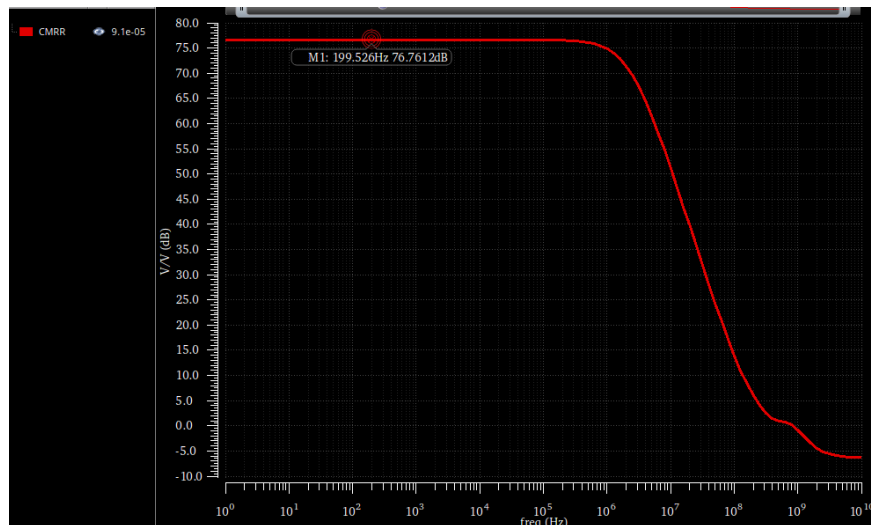


Figure 20: CMRR from XF (transfer-function) analysis: 76.76 dB at low frequency.

Table 5: CMRR comparison

Method	CMRR (dB)
Hand calc	74.0
$A_{vd} - A_{vcm}$ (subtracting Figs. 16, 19)	$73.11 - (-3.648) = 76.758$
XF analysis (Fig. 20)	76.76

The two independent simulation methods agree to two decimal places, and both clear the 74 dB spec by nearly 3 dB.

3.5 (Optional) Differential Large-Signal DC Sweep

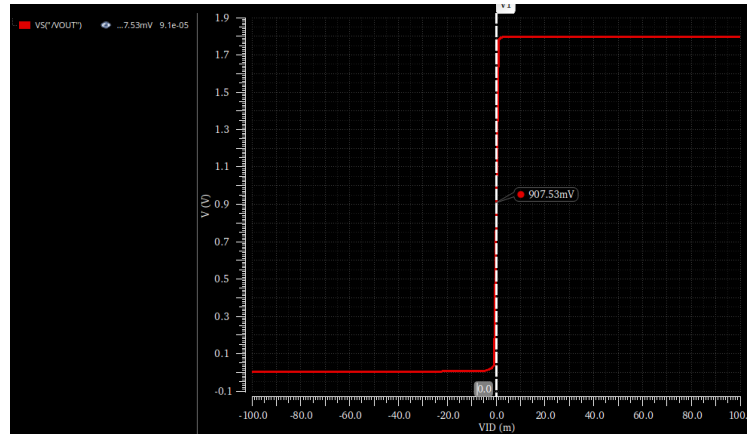


Figure 21: V_{OUT} vs. V_{ID} : at $V_{ID} = 0$, $V_{OUT} = 907.53$ mV.

Value at $V_{ID} = 0$ vs. DC OP: 907.53 mV vs. 907.62 mV from the DC-OP annotation – almost identical, confirming the test bench has zero external offset.

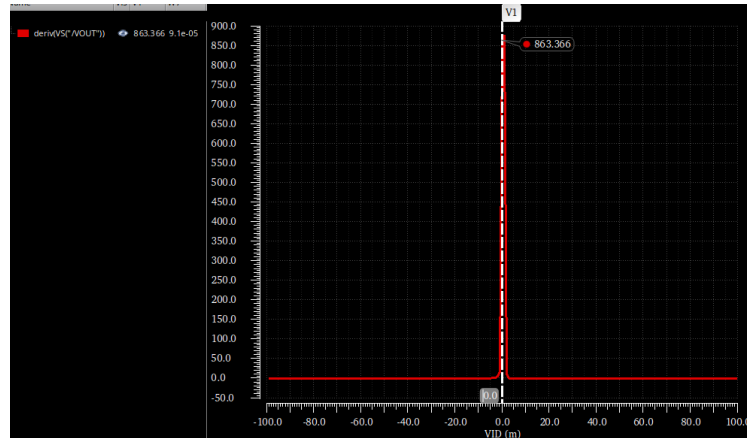


Figure 22: dV_{OUT}/dV_{ID} : peak = 863.4 V/V.

Is the peak less than A_{vd} from the AC analysis? Why?

A: Yes: $863.4 \text{ V/V} \ll 4525 \text{ V/V}$ ($= 73.11 \text{ dB}$). The step in V_{in} sweep is very large the true linear region width is only

$$\Delta V_{ID} \approx \frac{V_{DD}}{A_{vd}} = \frac{1.8}{4525} \approx 0.398 \text{ mV},$$

much smaller than the 1 mV sweep step. Using a much lower step ($\sim 1 \mu\text{V}$) shows the true 4525 V/V small-signal gain, as shown qualitatively by the finer-step re-run in Fig. 23.

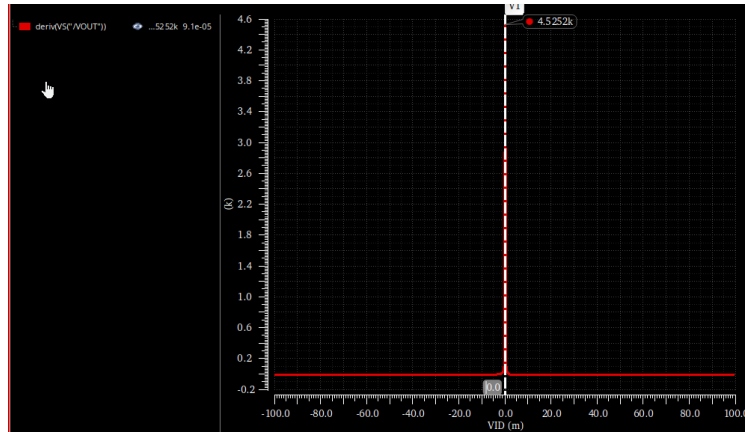


Figure 23: Derivative of V_{OUT} with a finer V_{ID} step, showing a taller, narrower peak closer to the AC small-signal gain.

3.6 Common-Mode Large-Signal DC Sweep (region vs. V_{ICM})

DC sweep, $V_{ICM} = 0$ to V_{DD} , plotting the **region** operating-point parameter for the input pair and the tail device.

First pass – CMIR-high fails

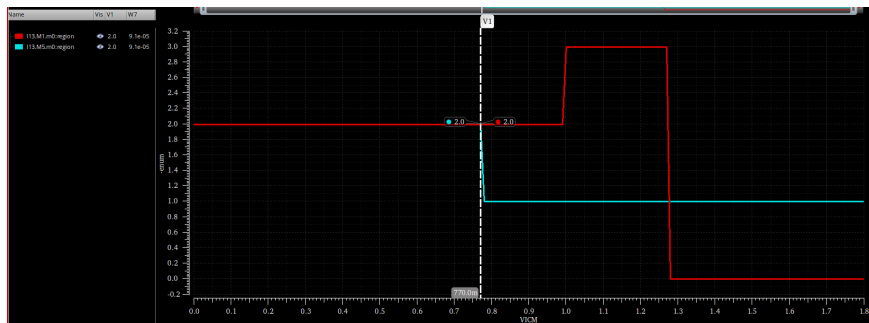


Figure 24: Region vs. V_{ICM} , initial sizing: M5 (tail) leaves saturation (region 2 $\rightarrow$ 1) at 770 mV.

Fix and re-check

Following the $V_{ICM,max} = V_{DD} - |V_{DSAT,tail}| - |V_{GS1}|$, headroom was bought back by increasing $|V_{GS1}|$'s drive, upsizing the input pair from $10.32 \mu\text{m}$ to $15 \mu\text{m}$ will fix that.

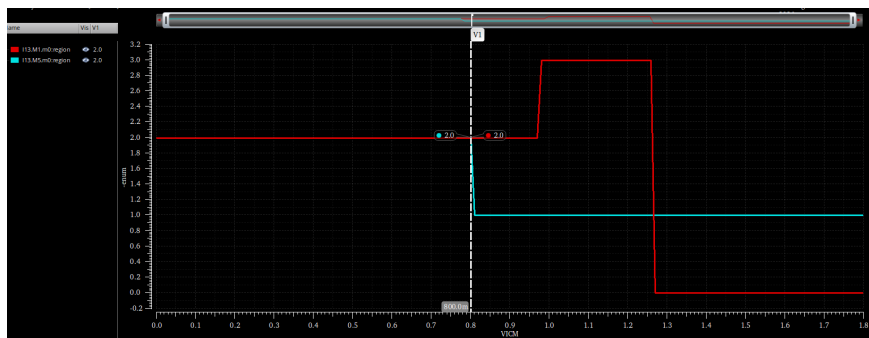


Figure 25: Region vs. V_{ICM} after upsizing M1/M2 to 15 μm : M5 now stays in saturation exactly up to 800 mV.

Extended (negative) sweep and CMIR-low

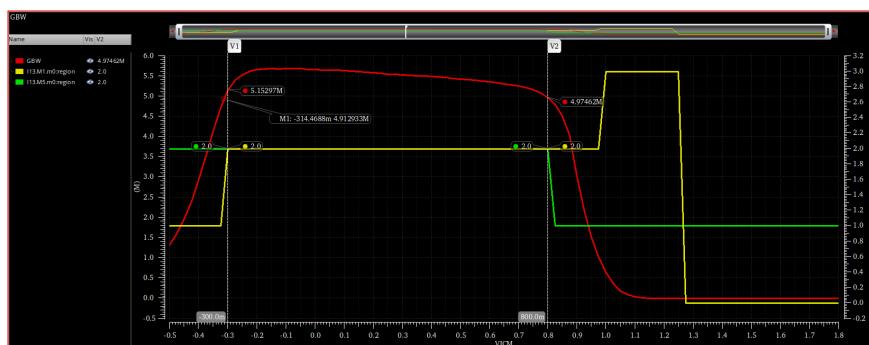


Figure 26: Extending the sweep below 0 V: M1 (input pair) stays in saturation until $V_{ICM} \approx -300\text{ mV}$.

Why can CMIR extend below ground for a PMOS input pair?

A: The drain of M1 (gate of M6) is pinned near $V_{GS3} \approx 744\text{ mV}$ by the diode-connected mirror load M3, essentially independent of V_{ICM} . As V_{ICM} (the gate of M1) is swept down – even below 0 V – M1’s source (the common tail node) follows it downward while $|V_{SG1}|$ stays roughly constant in saturation, so V_{SD1} (source-to-pinned-drain) actually *increases* as V_{ICM} drops, keeping M1 in saturation. M1 only leaves saturation once V_{ICM} has dropped so far that the condition for saturation ($V_{SD1} > |V_{ov1}|$) is violated, which the simulation places at $\approx -300\text{ mV}$. This confirms the CMIR-low $\leq 0.2\text{ V}$ spec with an big margin.

Table 6: CMIR: hand analysis vs. simulation

Parameter	Hand analysis (design target)	Simulation (region method)
CMIR-high	0.8 V	0.8 V
CMIR-low	$\leq 0.2\text{ V}$ (spec)	$\approx -0.3\text{ V}$

3.7 (Optional) Common-Mode Large-Signal GBW vs. V_{ICM}

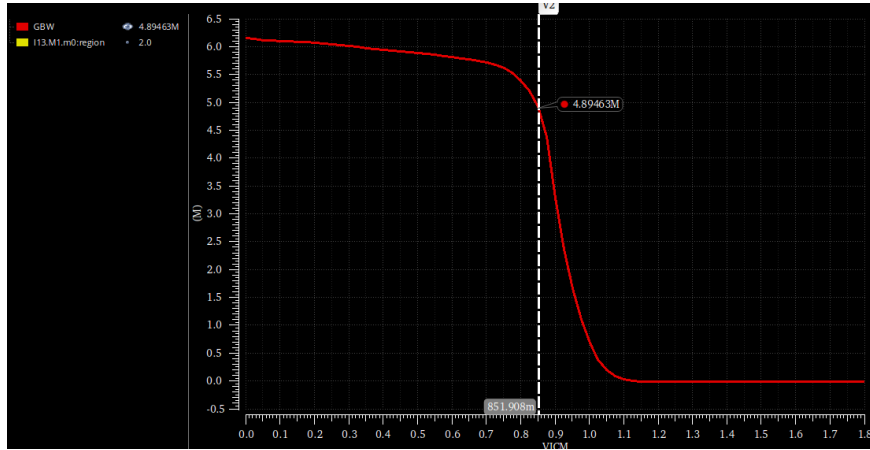
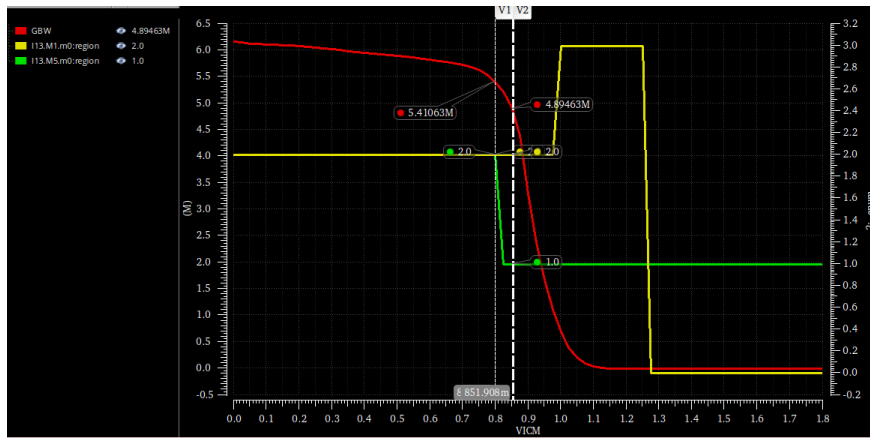
Figure 27: GBW vs. V_{ICM} .

Figure 28: GBW vs. V_{ICM} overlaid with the region-vs- V_{ICM} plot: the GBW collapse tracks the tail device (M5) leaving saturation exactly at 0.8 V, and the input pair (M1) staying saturated down to 0 V (and, per Fig. 26, below).

explanation: at low V_{ICM} , M5 has maximum headroom, so the tail current (and $g_{m1} \propto \sqrt{I_{tail}}$) is at its nominal value and GBW is maximum (≈ 6.15 MHz). As V_{ICM} approaches 0.8 V, M5 is pushed into triode, the tail current collapses, g_{m1} drops, and since $GBW \approx g_{m1}/(2\pi C_C)$, GBW gets smaller with it.

Table 7: Summary of Final Design (Part 3)

Design Component	Final Adjustment	Design Objective / Outcome
Current Mirror Load	$W = 6.12\mu$	balance the output at mid-rail.
Input Differential Pair	$W = 15\mu$	Improved CMIR-high headroom margins.
Compensation Capacitor (C_c)	Decreased value to 2.3 pF	Higher GBW

4 Part 4: Closed-Loop OTA Simulation

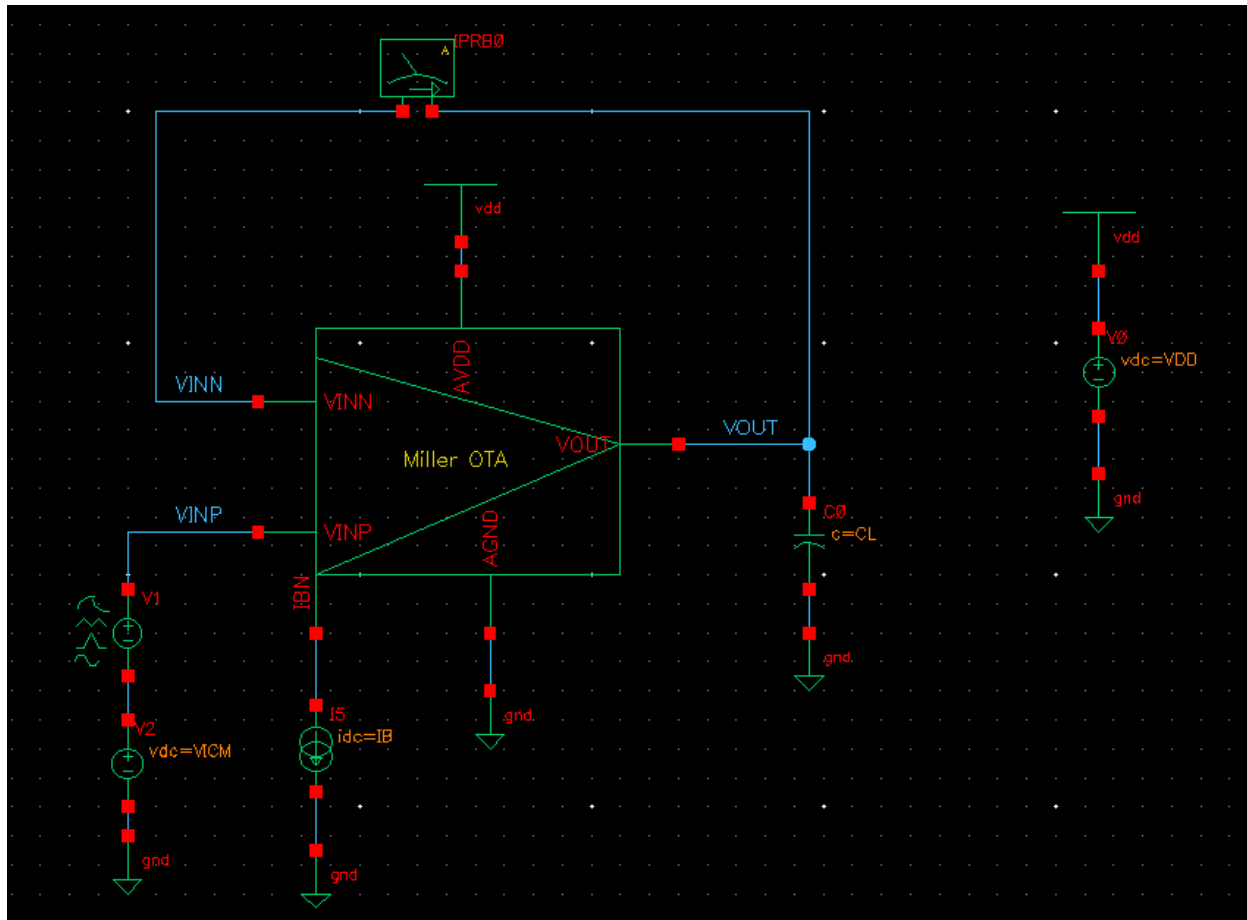


Figure 29: Closed-loop (unity-gain buffer) test bench with feedback-loop.

4.1 DC Operating Point

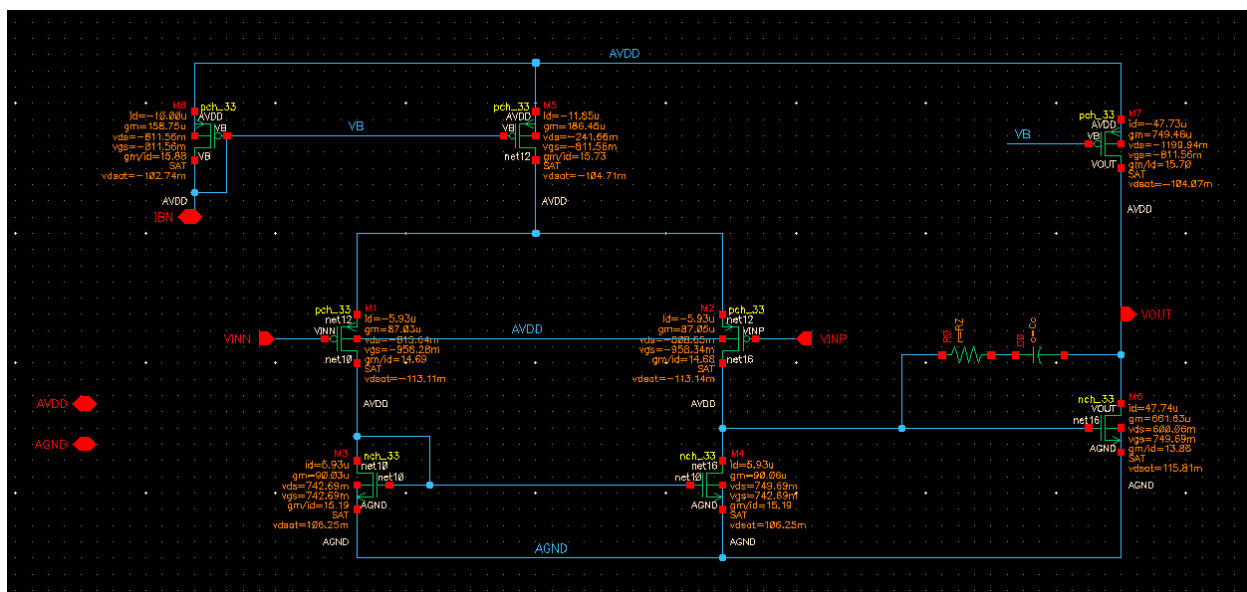


Figure 30: Closed-loop DC operating point, $V_{ICM} = V_{DD}/3 = 0.6 \text{ V}$.

Are V_{INP} and V_{INN} exactly equal? Why?

A: yes ($V_{INP} = 0.6\text{ V}$, $V_{INN} \approx 600.06\text{ mV}$, a $60\text{ }\mu\text{V}$ error). This is the virtual-short principle: with $A_{vOL} \approx 4916\text{ V/V}$ and negative feedback, $V_{INP} - V_{INN} = V_{OUT}/A_{vOL}$ is forced to be negligibly small.

Is the current (and g_m) in the input pair equal? Why?

A: Yes: $I_{D1} = I_{D2} = 5.93\text{ }\mu\text{A}$, $g_m \approx 87.0\text{ }\mu\text{S}$ each. Because feedback forces $V_{ID} \approx 0$, the tail current still splits 50/50.

Is the first-stage output DC voltage equal to the open-loop value? Why?

A: No: 749.69 mV here vs. 742.58 mV in open loop. In open loop, the offset-cancellation sweep centered V_{OUT} at $\approx 0.9\text{ V}$; in closed loop, feedback instead forces $V_{OUT} = V_{ICM} = 0.6\text{ V}$. Changing V_{DS6} from ≈ 0.9 to $\approx 0.6\text{ V}$ slightly changes the gate drive M6 needs to sink its $45\text{ }\mu\text{A}$ bias current (finite output resistance / channel-length modulation), which shifts the first-stage output node slightly.

4.2 Loop Gain (STB Analysis)

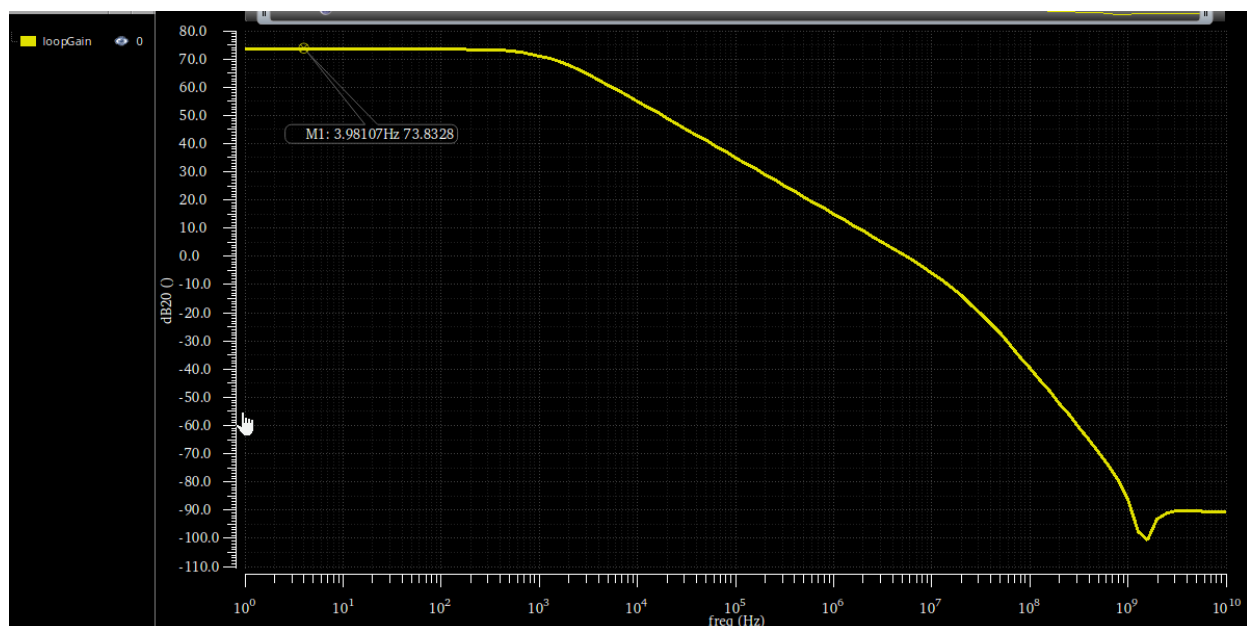


Figure 31: Loop gain magnitude (dB) vs. frequency (STB analysis).

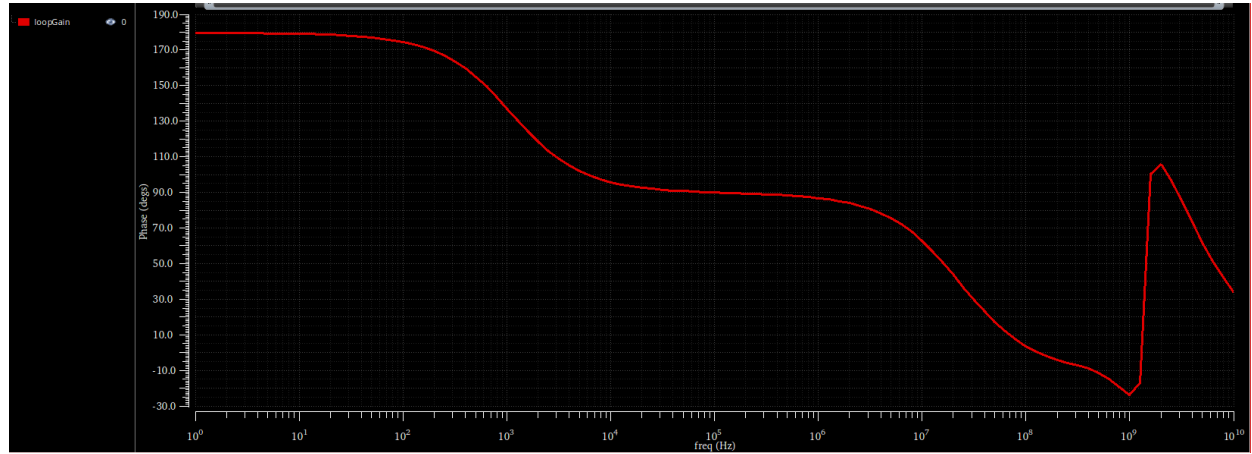


Figure 32: Loop gain phase vs. frequency, showing the high-frequency phase produced by R_Z pushing the RHP zero to the left-half plane.

ITI_LABS:miller_ota_tb_2:1	ymin(mag(getData("loopGain" ?...))	4.916k
ITI_LABS:miller_ota_tb_2:1	ymin(dB20(mag(getData("loop...))	73.83
ITI_LABS:miller_ota_tb_2:1	Phase Margin	75.32
ITI_LABS:miller_ota_tb_2:1	gainBwProd(mag(getData("loop...))	5.374M
ITI_LABS:miller_ota_tb_2:1	unityGainFreq(mag(getData("lo...))	5.23M

Figure 33: STB analysis summary marker: A_0 , f_u , PM.

Table 8: Open-loop AC vs. closed-loop STB comparison, and hand-calculated phase margin

Quantity	Open-loop AC	Closed-loop STB	Hand calc
A_0	73.11 dB	73.83 dB	≈ 72 dB
f_u	4.642 MHz (before changing C_c)	5.23 MHz	–
PM	–	75.32°	75.96°

The hand-calculated PM uses $PM \approx 90^\circ - \arctan(\omega_u/\omega_{p2})$ with the Step-10 design assumption $\omega_{p2} = 4\omega_u$:

$$PM \approx 90^\circ - \arctan(1/4) = 90^\circ - 14.04^\circ = 75.96^\circ$$

which matches the simulated 75.32° within 0.6° – excellent agreement, and comfortably clears the $\geq 70^\circ$ spec. The slight gain/bandwidth increase from open-loop to closed-loop is attributed to the different DC operating point (VOUT at 0.6 V instead of 0.9 V slightly changes g_{ds} and junction capacitances of M6/M7).

4.3 Slew Rate

Large-signal step applied to VINP: initial = CMIR-low +50 mV = 50 mV (using the practical, single-supply CMIR-low of 0 V rather than the extended -0.3 V limit), final = CMIR-high -50 mV = $0.8 - 0.05 = 750$ mV, delay 1 μ s, rise time 1 ns. Transient, 5 μ s stop, 0.1 ns max step.

Iteration 1: $C_C = 2.5$ pF (fails)

Test	Output	Nominal
ITI_LAB5:miller_ota_tb_2:1	slewRate(VT("/VOUT") 0.12 nil 0....	4.863M

Figure 34: Slew rate, $C_C = 2.5$ pF: 4.863 V/ μ s, just under the 5 V/ μ s spec.

Hand calc: $SR = I_{B1}/C_C = 11.79 \mu A / 2.5 \text{ pF} \approx 4.716 \text{ V}/\mu\text{s}$, close to the 4.863 V/ μ s.

Iteration 2: $C_C = 2.2$ pF (passes)

Since $SR = I_{B1}/C_C$, reducing C_C directly increases SR without touching the DC bias network. $C_C = 2.3$ pF was tried first (predicted $\approx 5.13 \text{ V}/\mu\text{s}$), then finalized at $C_C = 2.2$ pF:

Test	Output	Nominal
ITI_LAB5:miller_ota_tb_2:1	slewRate(VT("/VOUT") 0.12 nil 0....	5.058M

Figure 35: Slew rate, $C_C = 2.2$ pF: 5.058 V/ μ s – spec met.

Hand calc: $SR = I_{B1}/C_C = 11.8 \mu A / 2.2 \text{ pF} \approx 5.3 \text{ V}/\mu\text{s}$, close to the 4.863 V/ μ s.

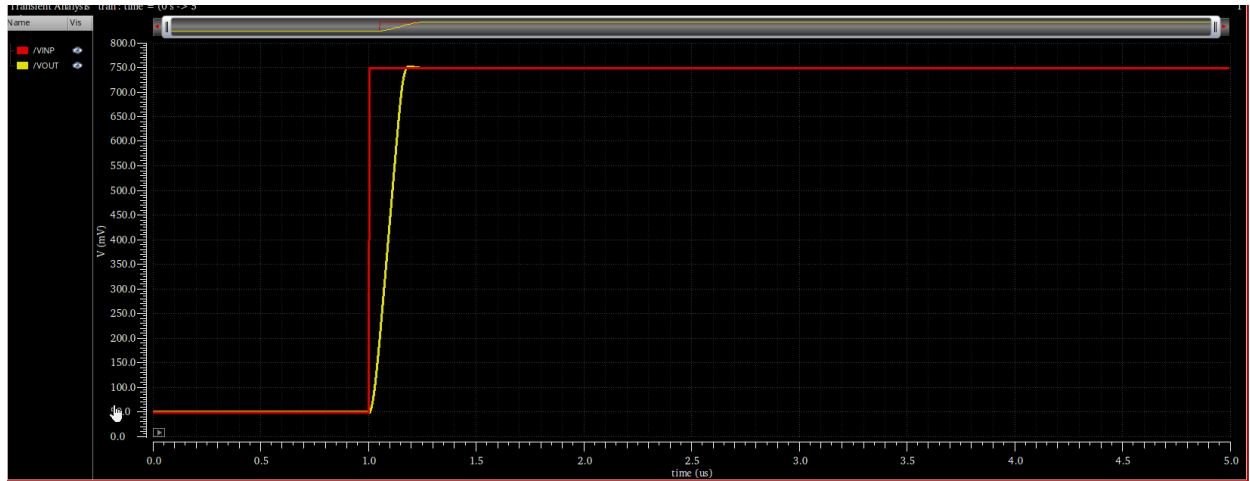
Figure 36: VIN and VOUT overlaid, $C_C = 2.2$ pF.

Table 9: Slew rate: hand calculation vs. simulation

C_C	Hand calc (I_{B1}/C_C , $I_{B1} = 11.79 \mu A$)	Simulation
2.5 pF	4.716 V/ μ s	4.863 V/ μ s
2.2 pF	5.359 V/ μ s	5.058 V/ μ s

Trade-off: reducing C_C raises ω_u , pushing it closer to ω_{p2} and slightly reducing phase margin.

4.4 Settling Time (Small-Signal Rise Time)

Small-signal step on VINP: initial = $V_{DD}/3 = 600 \text{ mV}$, final = $V_{DD}/3 + 5 \text{ mV} = 605 \text{ mV}$, same delay/rise-time/period as before.

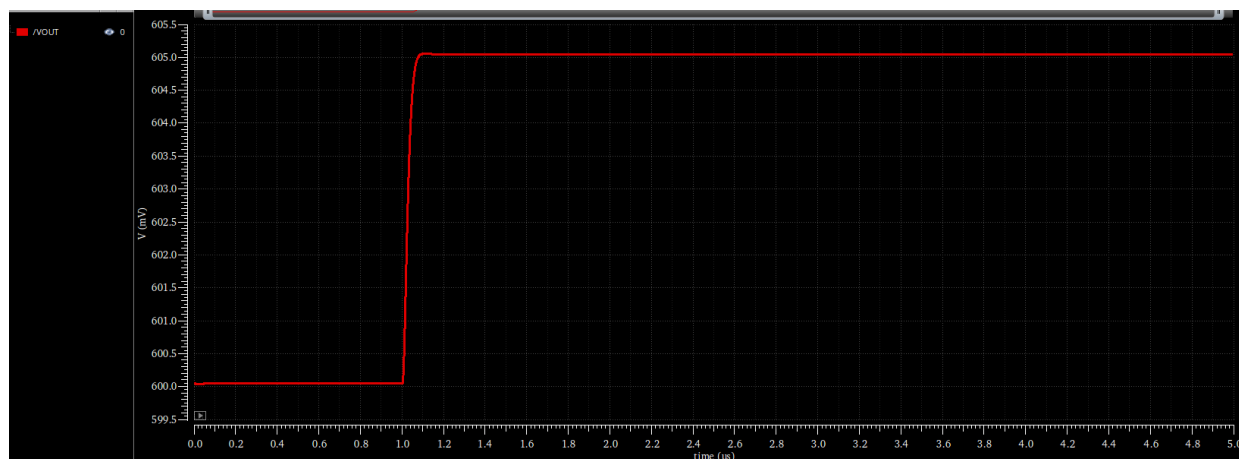


Figure 37: Small-signal step response: a clean settle to 605 mV with no visible overshoot or ringing.

Test	Output	Nominal
ITI_LABS:millar_ota_tb_2:1	riseTime(VT("VOUT")) 0.6 nll 0.6...	40.48 n

Figure 38: Rise-time measurement (10%–90%): 40.48 ns.

Hand calc: $t_{rise} = 2.2/\omega_u$, $\omega_u = g_{m1}/C_C = 87\mu S/2.2pF$, $t_{rise} = 55.6ns$

Table 10: Rise time: hand calculation vs. simulation

	Value
Hand calc	55.6 ns
Simulation	40.48 ns
Spec	≤ 70 ns

Do you see any ringing? Why?

A: No ringing. Ringing in the time domain is the direct signature of low phase margin in the frequency domain; PM below $\approx 60^\circ$ typically produces visible ringing, and PM below $\approx 45^\circ$ produces significant overshoot. Since PM = 75.32° here (critical damped system).

5 Part 5 (Optional): DC-Closed-Loop / AC-Open-Loop Simulation

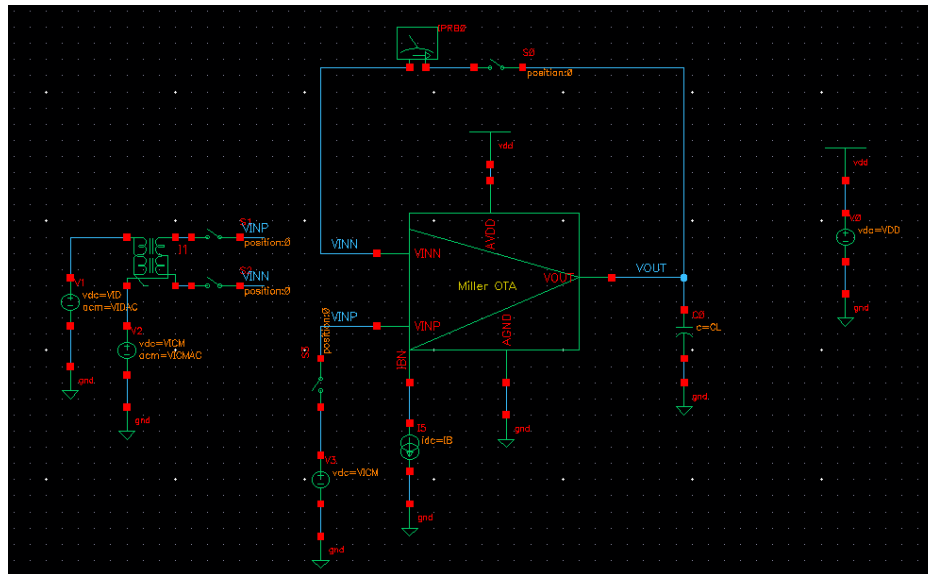


Figure 39: Switch-based DC-closed-loop / AC-open-loop test bench.

ITI_LABS:miller_ota_tb:1	VF("/VOUT")	
ITI_LABS:miller_ota_tb:1	dB20(VF("/VOUT"))	
ITI_LABS:miller_ota_tb:1	Ao	5.043k
ITI_LABS:miller_ota_tb:1	Ao_dB	74.05
ITI_LABS:miller_ota_tb:1	BW	1.063k
ITI_LABS:miller_ota_tb:1	fu	5.225M
ITI_LABS:miller_ota_tb:1	GBW	5.363M

Figure 40: AC response of the switch-based test bench.

Table 11: Part 5 AC results vs. Part 4 STB results

Quantity	Part 5 (switch-based AC)	Part 4 (STB)
A_0	74.05 dB	73.83 dB
f_u	5.225 MHz	5.23 MHz
GBW	5.363 MHz	5.374 MHz

6 Bonus - Total Area

Device	Type	W (μm)	L (μm)	Area (μm^2)	I_D (μA)
M1 (Input)	PMOS	15.00	0.52	7.800	5.93
M2 (Input)	PMOS	15.00	0.52	7.800	5.93
M3 (Load)	NMOS	6.10	0.67	4.087	5.93
M4 (Load)	NMOS	6.10	0.67	4.087	5.93
M5 (Tail)	PMOS	25.36	0.34	8.622	11.85
M6 (Driver)	NMOS	22.96	0.40	9.184	45.46
M7 (Driver Load)	PMOS	91.00	0.34	30.940	45.46

Table 12: Active transistor area and DC current breakdown.

$$Area_{total} = 7.8 + 7.8 + 4.087 + 4.087 + 8.622 + 9.184 + 30.94 = 72.52\mu\text{m}^2$$

2. Area $\times$ Current Product

To find the total Area-Current product, we sum the product of each transistor's area and its respective DC drain current:

$$\text{Area-Current Product} = \sum_{i=1}^{12} (Area_i \times I_{D,i})$$

Breaking this down by device (in $\mu\text{m}^2 \cdot \mu\text{A}$):

- **M1, M2:** $2 \times (7.800 \times 5.93) = 92.51$
- **M3, M4:** $2 \times (4.087 \times 5.93) = 48.47$
- **M5:** $8.622 \times 11.85 = 102.17$
- **M6:** $9.184 \times 45.46 = 417.50$
- **M7:** $30.940 \times 45.46 = 1406.53$

Summing these individual products yields:

$$\text{Total Area} \times \text{Current} = 2067.18 \mu\text{m}^2 \cdot \mu\text{A}$$

without summing:

$$\text{Total Area} \times \text{Total Current} = 72.52 * 57 = 4133.64 \mu\text{m}^2 \cdot \mu\text{A}$$

Part 6 (Bonus): PVT-Independent Compensation

Design Procedure and Hand Analysis

To make our lead compensation independent of process and temperature variations, we replaced the ideal nulling resistor with an NMOS transistor (M_9) operating in the triode region. The goal is to make the resistance of M_9 (R_Z) track the inverse transconductance of our second-stage driver ($1/g_{m6}$).

To do this, we added a new bias branch (M_{10} , M_{11} , and M_{12}) to generate the gate voltage (V_c) for M_9 .

1. Matching the Current Density:

For the tracking to work, the bottom diode (M_{12}) must have the same effective gate-source voltage as the driver (M_6). We achieve this by keeping their current densities equal:

$$\frac{I_{D6}}{I_{D12}} = \frac{(W/L)_6}{(W/L)_{12}}$$

From our DC operating point simulation, M_6 draws $45.46 \mu\text{A}$ and our new bias branch draws $4.13 \mu\text{A}$. The current ratio is $45.46/4.13 \approx 11$. To maintain the same density, we set the width of M_{12} to be roughly 1/11 of the width of M_6 .

2. Sizing the Triode Resistor:

$$R_Z g_{m6} = \frac{(W/L)_6}{(W/L)_9} \sqrt{\frac{(W/L)_{11}}{(W/L)_{12}}}$$

Since we want to place our zero at infinity to cancel the non-dominant pole, we need $R_Z = 1/g_{m6}$, which means $R_Z g_{m6} = 1$.

This simplifies our design equation to:

$$W_9 = W_6 \sqrt{\frac{W_{11}}{W_{12}}}$$

Simulation Results

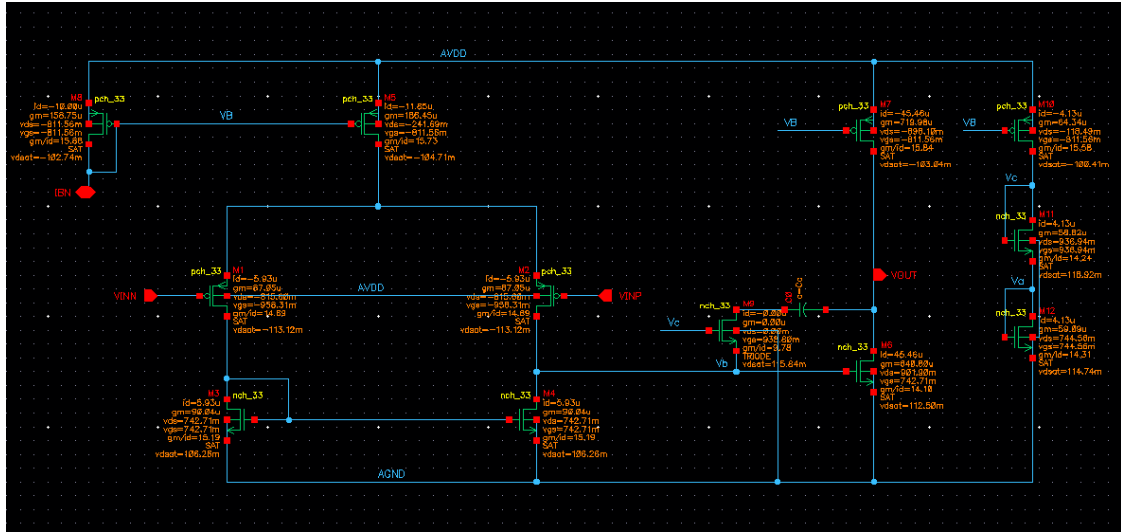


Figure 41: Schematic of the PVT-independent compensation network with DC operating points.

Looking at the DC annotations, M_9 is successfully operating in the TRIODE region. To verify the tracking accuracy, we extracted the channel resistance (r_{on}) of M_9 and compared it to $1/g_{m6}$.

ITL_LABS:test1	pv("I0.M9.m0" "ron" ?result "dc...)	1.64k
ITL_LABS:test1	(1 / pv("I0.M6.m0" "gm" ?result "...)	1.561 k

Figure 42: Simulated R_Z vs. theoretical $1/g_{m6}$ target.

The simulated R_Z is **1.64 kΩ**, which tracks with the theoretical $1/g_{m6}$ value of **1.561 kΩ**.

PVT Corner Analysis & Headroom Limitations

Parameter	Nominal							C0_0	C0_1	C0_2
Save_All_OP...										
VICM	600m							600m	600m	600m
diodes.scs	tt							tt	tt	tt
finfets.scs	tt							tt	tt	tt
mim_cap.scs	tt							tt	tt	tt
mosfetb.scs	tt							ff	tt	ss

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C0_0	C0_1	C0_2
ITI_LABSmiller_ota_tb_comp.2:1	RZ	1.639k				1.3k	20.21k	1.3k	1.639k	20.21k
ITI_LABSmiller_ota_tb_comp.2:1	1/gm6	1.512k				1.43k	1.596k	1.43k	1.512k	1.596k

Figure 43: Corner simulation showing R_Z tracking across tt, ff, and ss corners.

We ran a PVT corner simulation to verify the robustness of the tracking circuit.

- In the **tt (Typical)** and **ff (Fast-Fast)** corners, the tracking works perfectly (1.639 k Ω and 1.3 k Ω respectively).
- However, in the **ss (Slow-Slow)** corner, R_Z spikes massively to **20.21 k Ω** .

Why does it fail in SS?

the threshold voltages (V_{th}) increase. The stacked diodes (M_{11} and M_{12}) require a much larger V_{GS} to sink their 4.13 μA current. This pushes the voltage at node V_c so high that it slams into our 1.8V supply rail, crushing the PMOS current source (M_{10}) into the triode region. Once M_{10} is starved of current, the bias voltage drops, and M_9 barely turns on, causing its resistance to increase.

7 Summary of Design

- (1) **Input pair (M1/M2) width increased from 10.32 μm to 15 μm** (Section 3.6), to reduce $|V_{GS1}|$ and get more headroom for the tail device M5.
- (2) **Mirror-load (M3/M4) width swept from 6.36 μm down to 6.1 μm** to cancel the systematic offset that was initially pushing V_{OUT} up to 1.289 V instead of ≈ 0.9 V.
- (3) **Compensation capacitor C_C reduced in two steps, 2.5 pF $\rightarrow$ 2.3 pF (trial) $\rightarrow$ 2.2 pF (final)**, to raise the slew rate from a failing 4.863 V/ μs to a passing 5.058 V/ μs , at the cost of a small (acceptable) reduction in phase margin.

Table 13: Final specification compliance summary

Spec	Target	Achieved	Status
Static gain error	$\leq 0.05\%$ ($A_{vOL} \geq 66$ dB)	73.83 dB (STB)	✓
CMRR	≥ 74 dB	76.76 dB	✓
Phase margin	$\geq 70^\circ$	75.32 $^\circ$	✓
Total current	≤ 60 μA	57.21 μA	✓
CMIR-low	≤ 0.2 V	≈ -0.3 V	✓
CMIR-high	≥ 0.8 V	0.8 V	✓
Output swing	0.2 V to 1.6 V	Verified in Part 3 (V^* margins)	✓
Closed-loop rise time	≤ 70 ns	40.48 ns	✓
Slew rate	≥ 5 V/ μs	5.058 V/ μs	✓